

Sanyu Studio

A Multi-Agent System for Art-Historical Narrative Construction

Zhaoxi Wei

Renmin University of China, weizhaoxi25@ruc.edu.cn

Hongye Yang

Georgia Institute of Technology, hyang783@gatech.edu

Shuyuan Tian

University of Cologne (Universität zu Köln), stian@smail.uni-koeln.de

Amid concerns that generative AI may standardize art interpretation, this paper examines whether LLM-based interaction can support plural art-historical narrative construction. We present Sanyu Studio, a multi-agent dialogue system that models 321 Sanyu oil paintings as agents with fact, interpretation, organization, and memory-filtering mechanisms. Based on a seven-day workshop with eight art-university participants, the study shows that user prompts, evidence organization, and cognitive tendencies shaped divergent yet coherent versions of “digital Sanyu.” The findings suggest that, under conditions of limited historical evidence, AI can amplify human agency and offer public audiences an interactive entry point into art-historical interpretation.

CCS CONCEPTS • Applied computing • Arts and humanities • Fine arts

Additional Keywords and Phrases: LLM, Art Interpretation, HCI, Art history writing, Sanyu,

1 INTRODUCTION

Art-historical narrative is increasingly mediated by LLMs, which now shape how readers access and interpret art-historical texts [15]. Traditional interpretation depends on the interplay between verifiable materials and scholarly perspectives: images, exhibition records, letters, and interviews provide evidence, while iconography, social history, and psychoanalysis shape how such evidence is organized and interpreted [8, 12]. This shift raises concern that LLM-based interpretation may homogenize and flatten artistic meaning through statistical generation [1].

This paper argues that a designed multi-agent LLM system can simulate the formation of art-historical narrative. External knowledge bases and retrieved materials provide evidence; LLMs organize evidence, connect relations, and generate interpretations; and users’ questions continually redirect the narrative entry point. Under constrained system design, AI can function as a narrative mechanism for nonexpert researchers, enabling more plural forms of artistic meaning.

Existing work on LLM-based art interpretation falls into three directions. First, multimodal models analyze painting features and cross-cultural meanings [3, 19], but remain limited to static image interpretation and lack multi-work narrative

design. Second, RAG and external knowledge bases supplement art-historical context [9, 14], yet do not address the long-tail distribution of art-historical materials. Both approaches focus on static knowledge retrieval and overlook how user interaction reshapes interpretive paths. Third, historical-figure chatbots support museum and heritage education [4, 10], but fixed personas limit their ability to generate open-ended biographical narratives from diverse materials and questions. How LLM systems can simulate art-historical narrative formation therefore remains underexplored.

In response, this paper presents Sanyu Studio, a multi-agent art-historical narrative system centered on Sanyu. Sanyu is selected because, although his works have gained visibility through exhibitions and auctions, his artistic persona continues to be rewritten in later scholarship [18]. This openness makes his case suitable for examining how narrative emerges across works, evidence organization, and user questions. Sanyu Studio maps research questions, evidence retrieval, material negotiation, and narrative synthesis onto an interactive multi-agent mechanism, making the formation of art-historical narrative visible through questioning, selection, interpretation, and synthesis.



Figure 1: Conceptual image. As a globally recognized Asian artist, Sanyu remains an open subject of art-historical discussion because his biographical records are fragmented, and many interpretations of his works still depend on partial archival evidence.

2 METHODOLOGY

This section explains how Sanyu Studio proceduralizes art-historical narrative and lowers access barriers. User questions function as research questions; painting agents serve as retrieved evidentiary standpoints; roundtable discussion stages negotiation among materials; and the Sanyu agent records, filters, and synthesizes these exchanges, ultimately producing a continuously revisable image of Sanyu (Figure 2).

2.1 User Questions as Research Questions

In Sanyu Studio, user questions are framed as research questions. Users ask questions about Sanyu through the PC interface, thereby entering the formation of art-historical narrative. Each question determines which works the system retrieves and initiates the subsequent roundtable discussion (Figure 2).

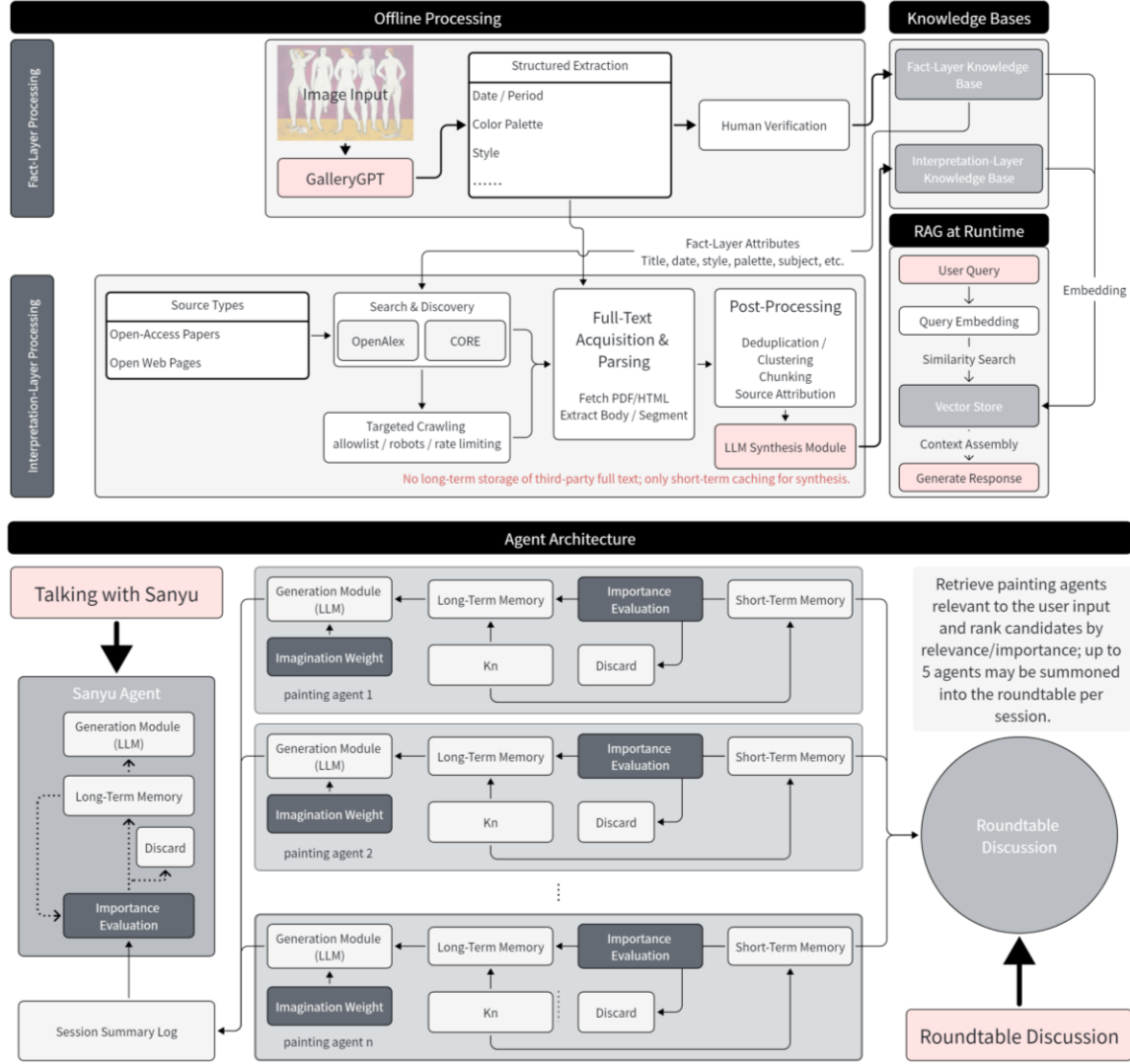


Figure 2: System architecture of Sanyu Studio.

2.2 Painting Agents as Evidentiary Positions

We model 321 Sanyu oil paintings, licensed from the XXX Foundation’s public catalogues and high-resolution image archive, as separate painting agents, each with its own knowledge base. After a user submits a question, the system retrieves relevant agents for roundtable discussion. Each agent’s response is constrained by its work metadata, visual features, and related literature, making it a distinct evidentiary source.

Each knowledge base has a factual layer and an interpretive layer. The factual layer, compiled from public catalogues, artwork archives, and verifiable exhibition records, records objective data such as date, size, and exhibition history. It also uses high-resolution images and GalleryGPT [3] to describe composition, color, and style. The interpretive layer retrieves

open-access scholarship, criticism, and art-historical essays, then uses Gemini 2.5 Flash-Lite to deduplicate content and extract viewpoints. The system stores only compressed interpretive entries and minimal source metadata, with all entries manually checked against the original image and source.

At runtime, the user question is vectorized, and each candidate agent retrieves the top- $k = 5$ factual and interpretive entries as context. Factual claims must come from the factual layer; interpretive content may include explicitly labeled, DSBM-weighted inferences. Thus, each painting agent provides an independent evidentiary position while preserving controlled interpretive space.

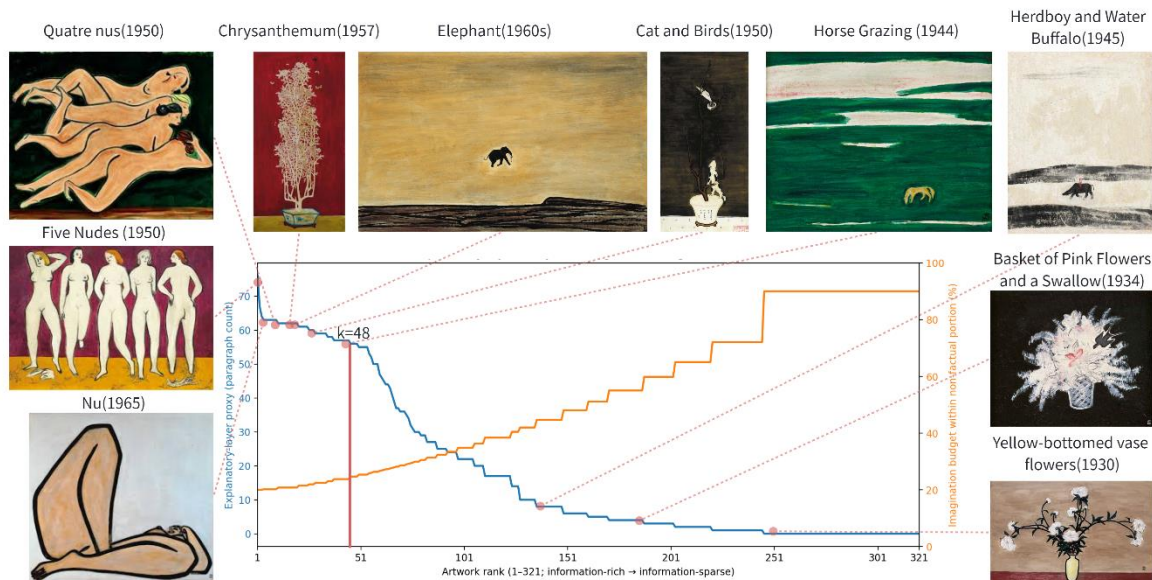


Figure 3: DSBM Evidence Distribution. The blue curve shows the explanatory-layer proxy, measured by deduplicated interpretive passages across 321 works. The orange curve shows the DSBM-derived speculative-inference budget. The distribution is long-tailed: the top $k = 48$ works account for 50% of all interpretive text.

2.3 Roundtable discussion as Interpretive Negotiation

The “roundtable discussion” coordinates multiple painting agents in Sanyu Studio (Figure 4). After a user submits a question, the system vectorizes it and retrieves relevant works from all agents. Candidate works are first ranked by relevance, then reranked through the document-scarcity balancing mechanism (DSBM). Up to five agents enter the discussion. This simulates how art-historical research reorganizes materials around a question by placing well-documented works alongside under-documented but interpretively valuable ones, forming a more open evidentiary structure [6, 17].

Before agent construction, the system measures the interpretive richness of 321 works using the number of deduplicated interpretive passages, p_i , as a proxy. The results show a clear long-tail distribution: about 15% of works contribute about 50% of the interpretive text, while the remaining 85% lack systematic discussion. Without adjustment, the roundtable would repeatedly center on high-information works. DSBM therefore allows under-documented works to enter evidentiary reasoning through visual details and comparative relations [7] (Figure 3).

To model this process, DSBM first computes the normalized information richness of work i :

$$z_i = \text{clip} \left(\frac{\log(1 + p_i) - Q_{0.05}}{Q_{0.95} - Q_{0.05} + \epsilon}, 0, 1 \right)$$

Here, $Q_{0.05}$ and $Q_{0.95}$ are the 5th and 95th percentiles of the $\log(1 + p_i)$ distribution across all works, and ϵ is a small positive constant. z_i denotes normalized information richness, while $1 - z_i$ denotes document scarcity. The system then computes the scheduling weight of work i in round t , given question q and selected set S :

$$\omega_i^{(t)}(q, S) = g_i \cdot [r_i(q)^\alpha + \beta(1 - z_i)^\gamma] \cdot [1 - D(e_i, S)]^\delta \cdot \exp(-\eta a_i^{(t-1)})$$

The cognitive-diversity penalty $D(e_i, S)$ is defined as:

$$D(e_i, S) = \begin{cases} 0, & S = \emptyset \\ \text{clip} \left(\max_{j \in S} \cos(e_i, e_j), 0, 1 \right), & S \neq \emptyset \end{cases}$$

Here, $r_i(q)$ denotes question relevance, $g_i \in [0, 1]$ is the evidence gate, and e_i is the vector representation of the work's knowledge base. $a_i^{t-1} = u_i^{t-1} / (\sum_k u_k^{t-1} + \epsilon)$ denotes normalized token-level attention, where u_i^{t-1} is the cumulative number of tokens assigned to work i before round t . $\alpha, \beta, \gamma, \delta, \eta$ are nonnegative hyperparameters. $D(e_i, S)$ measures the maximum cosine similarity between e_i and the selected set S , clipped to $[0, 1]$. When $S = \emptyset$, $D(e_i, S) = 0$.

DSBM also regulates agent speech. Less-documented works may offer explicitly labeled speculative interpretations under factual constraints, while well-documented works rely mainly on existing literature and verifiable information. Since art-historical inference often depends on visual traces and comparison, all speculation is labeled to avoid confusion with factual claims [7].

Once selected, each painting agent retrieves the top- $k = 5$ factual and interpretive entries from its own knowledge base as context. Gemini 2.5 Flash-Lite then generates responses with temperature set to 0.2 to reduce hallucination. Agents speak in the first person and refer to Sanyu in the third person, preserving the artwork–artist relation.

With DSBM enabled, the token-level speaking share of high-information works decreased from 41.8% to 20.4%, while the number of distinct works discussed increased from 151 to 279. This shows that DSBM broadens roundtable participation, giving under-documented works greater narrative agency. The roundtable thus models art-historical negotiation, where multiple materials complement and constrain one another under a shared question.

2.4 Memory Filtering as Narrative Stabilization

Historical narrative depends on the selective organization of facts and meanings [16]. Accordingly, Sanyu Studio does not retain all roundtable discussion. Each painting agent has short-term and long-term memory: short-term memory stores the latest 6–8 turns and appends them to the prompt, while long-term memory stores up to 100 high-importance fragments across sessions.

After each round, the system segments the dialogue into candidate fragments and uses an LLM to assign each an importance score from 1 to 10 based on the research goal and retrieval context. Scoring considers relevance to the current question, new factual or interpretive value, and connection to existing memory. Fragments scoring ≥ 7 are written into long-term memory; the rest are discarded. When the token budget is reached, the system merges recent high-scoring fragments and removes redundancy. Long-term memory stores interpretive paths, question trajectories, and narrative tendencies, while factual claims remain constrained by the fixed knowledge base.

1. Homepage

The homepage provides two entry points: “Roundtable discussion” and “Talking with Sanyu.” The latter remains locked and is unlocked only after the system has recorded a certain number of completed roundtable sessions.



2. “Roundtable discussion” page

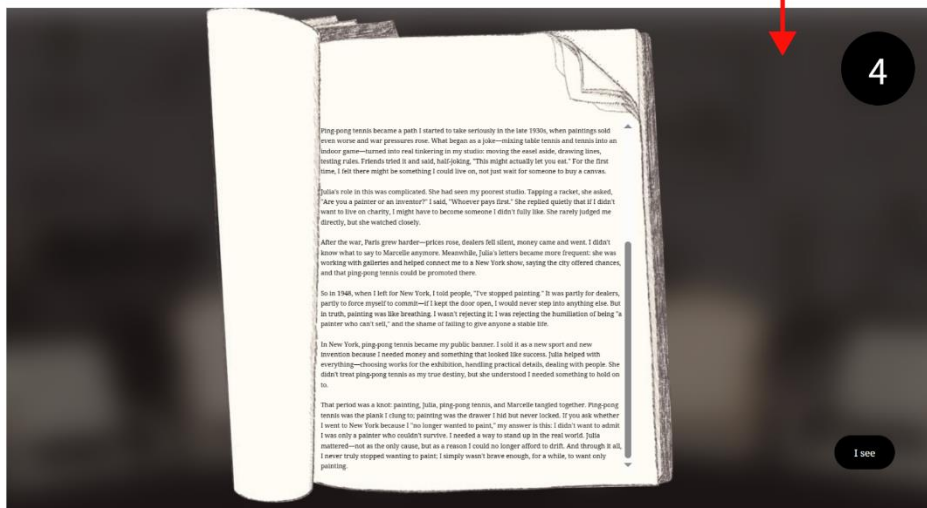
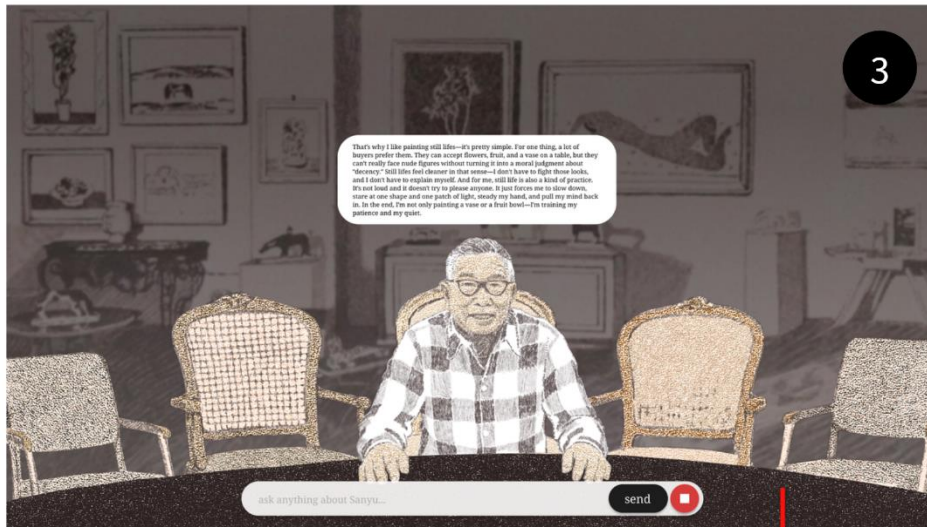
This page allows users to interact directly with painting agents that represent Sanyu’s individual works. The interaction flow is organized by the imagination mechanism and the importance mechanism.

Figure 4: Homepage and roundtable discussion interface of Sanyu Studio. The homepage provides two entry points: “Roundtable Discussion” and “Talking with Sanyu.” In the roundtable page, users interact with painting agents that represent individual works.

This converts the selection, compression, and organization of art-historical writing into a computable filtering mechanism, allowing selected questions, evidence relations, and interpretive frames to persist across interactions and form a stable yet revisable artist narrative [5].

3. "Talking with Sanyu" page

This page enables users to interact with Sanyu directly in a one-on-one conversation. The Sanyu agent answers based on its stored long-term memory, returning information that has been accumulated across prior sessions.



4. Sanyu's post-roundtable summary

After each roundtable session ends, the Sanyu agent generates a summary of the discussion. "Talking with Sanyu" does not trigger this step.

Figure 5: "Talking with Sanyu" interface and post-roundtable summary. Users can converse directly with the Sanyu agent, which responds based on accumulated memory from previous roundtable sessions. After each roundtable, the system generates a summary to update Sanyu's long-term memory.

2.5 Sanyu Agent as Narrative Synthesis

In art history, an artist's image is shaped through the repeated organization of works, texts, biography, and later scholarship [8]. In Sanyu Studio, after each roundtable, participating painting agents summarize the dialogue and long-term memory, then send these summaries to the Sanyu agent, which organizes them into narrative memory about Sanyu.

The Sanyu agent uses the same importance evaluation, long-term memory, and consolidation mechanisms to store, compress, and reorganize interpretive clues and narrative relations, while factual claims remain constrained by the fixed knowledge base. After ten roundtable rounds, users can enter “Dialogue with Sanyu” mode and question the Sanyu agent directly (Figure 5). If no relevant memory exists, the system states this and guides users to start a new roundtable.

Thus, user questions, agent selection, roundtable negotiation, and memory filtering converge in the Sanyu agent, translating dispersed materials into a revisable art-historical figure.

A

- 1. In Paris, the people who really kept helping me were a small frame-shop owner in Montparnasse named Madeleine, a Polish buddy called Janusz who kept covering my rent, and “Julia,” who sometimes quietly slipped my paintings into the corner of a salon.
- 2. Yeah—I rolled up my most important nudes and a few big works on paper in oilcloth, left them with a church caretaker called Father Leroy and a storage company that never asked questions, and I kept the only key sewn inside my coat lining.
- 3. The real deal was simple: Roché gave me a small monthly advance that basically kept me alive, and in return he got control over a batch of my paintings; if anything sold, he’d take costs first and then we’d split it “him 60, me 40,” but most years nothing sold and it was mostly just debt on paper.
- 4. The mirrors were old ones I picked up at flea markets and from a stage-props shop; I’d “name” the surface with a super thin ground first, then push and scrape the oil paint again and again—and I chose mirrors because I wanted viewers to see themselves first, and only then see my line.
- 5. Yes, it was a serious pivot—at least in New York I spent about half my time pitching and demoing ping-tennis—but I also knew it was a kind of desperate, respectable performance to save myself.

B

- 1. In Paris, the people who consistently supported me weren’t from the art world, but rather a Russian laundromat owner named “Inna,” a post office clerk named “Andre” who helped me write letters in French, and a baker named “Pierre” who left me half a loaf of bread every morning at dawn.
- 2. Yes, I entrusted a few of my most precious works to a so-called “Time Bank” club that only accepts art, not money, where I wrote the vault number on the last page of my sketchbook—but later I lost that sketchbook.
- 3. We never had any elegant contracts, just a piece of paper stained with wine, where he promised to arrange two public appearances for me and shield me from my debts, while I agreed to give him the works I could sell first, as for the money, most of it was already taken by Parisian landlords and taverns.
- 4. The mirror wasn’t something I found; it was a stage makeup artist named “Marcel” who used it as a practice board for light reflection, and I painted on it because I refused to absorb the paint, forcing me to treat each stroke as a one-way gamble.
- 5. It wasn’t a career shift; I just wanted to make poverty sound more dignified: what I was selling wasn’t rackets or rules, but the possibility that “I could still survive.”

C

- 1. In Paris, I had a close relationship with a gallery owner named “Arno,” who was not only my art supporter but also provided financial backing for me.
- 2. I entrusted a few of my important works to my friend “Franz Bader,” who has been keeping them safe ever since.
- 3. My collaboration with Henri-Pierre Roché was never formalized, it was more of a verbal agreement, but he did provide me with significant financial help, although he wasn’t involved in the creative process.
- 4. I chose the mirror as a medium because it reflects the surrounding environment, creating a kind of “mirror image” in my artwork, which at the time symbolized my exploration of the line between reality and illusion.
- 5. Actually, my reason for promoting “ping-tennis” in New York wasn’t a career shift, but I was inspired by an unknown sports promoter named “Will Forman,” who sparked my interest and creativity in this sport.

D

- 1. In Paris, the person I trusted most was “Isabelle Montal,” who not only was my close friend but also provided me with continuous inspiration and financial support for my art.
- 2. My important works were entrusted to the “Bernard Gallery” in Paris, but this wasn’t a formal agreement; it was based on trust and long-term friendship.
- 3. My collaboration with Henri-Pierre Roché was mostly about business, where we organized exhibitions together, and I received favorable contracts from him, but he didn’t participate directly in the creation of my works.
- 4. I got the mirror material through a private channel and used a special brush and metallic paint to ensure a clear reflective effect. I chose this material because it allowed me to incorporate the viewer’s reflection into the artwork.
- 5. When I promoted “ping-tennis” in New York, it wasn’t part of a career change. Instead, I was inspired by “George Winston,” a little-known sports advocate, who encouraged me to combine this sport with my artistic work.

E

- 1. In Paris, my closest friend was “Jacques Dubois,” who was not only my gallery partner but also provided significant emotional and financial support when I faced creative struggles.
- 2. A few of my key works were entrusted to the “Monte Carlo Art Foundation,” which has been safeguarding them, though this relationship never had an official written agreement.
- 3. My collaboration with Henri-Pierre Roché was more of an ongoing art experiment; there wasn’t much financial exchange, but he did provide a venue for one of my exhibitions in New York.
- 4. I received the mirror material from a Swiss merchant named “Pierre Gill,” who showed me how to paint on it and ensured the effect was clear. I chose this material because it gave my artwork a stronger sense of reality and a dream-like quality.
- 5. When I promoted “ping-tennis” in New York, I wasn’t seriously considering a career shift. Instead, I was inspired by “Fred Hart,” a museum scholar, who believed this sport could bring a unique perspective to my artistic work.

F

- 1. In Paris, my most trusted friend was “Adolphe Proune,” who was an art critic and provided both creative space and necessary financial support for me.
- 2. A few of my important works were entrusted to the “Paris Art Bank,” where they’ve been safely kept, and they promised to give me exhibition opportunities whenever needed.
- 3. My relationship with Henri-Pierre Roché was mainly based on artistic understanding. He occasionally provided exhibition space for my works but never got directly involved in the transactions.
- 4. I obtained the mirror material through an Italian supplier named “Fabio Vittoni,” who showed me how to use special paints for the work and emphasized the space and time effects created by the reflective surface.
- 5. My promotion of “ping-tennis” in New York wasn’t a career shift; I was inspired by “Claude Porter,” a little-known New York sports promoter, who showed me how to blend sports and art in a unique way.

G

- Back in Paris, the ones who really kept me afloat were a quiet landlord called “Madame Violette,” a Moroccan café waiter “Samir,” and a broke violin student “Luc,” who still shared his dinner with me.
- Yeah—I left a few pieces with a tiny photo studio near Montparnasse; the owner stamped my name on the back of the frames and told me, “Come back when you’re ready.”
- With Roché, it was blunt: he advanced me cash when I was desperate, and in return he got first pick of whatever might sell—no romance, just survival math.
- I got the mirror sheets from a demolished dance hall, sealed them with a thin coat of varnish, then painted fast with small strokes and wiped-back highlights, because I wanted the surface to fight me and stay alive.
- Yes, I meant it—I pushed ping-tennis like it was my second life, not a hobby, even if part of me knew it was a long shot.

H

- 1. The real ones in Paris were a shy translator “Claire,” a Czech carpenter “Marek,” and a nun from a small charity kitchen—none of them cared about my fame, only whether I ate.
- 2. Yes—I locked a few works into a shipping company’s storage locker; the clerk wrote the code on a matchbox, and that’s how Roché, the truth is messy: he wanted a clean artist story, I wanted cash and distance, so we kept renegotiating until both of us got tired.
- 4. I got the mirror from a theatre props warehouse, fixed it with shellac and glue sizing, then painted in thin layers and scratched through them, because I wanted the image to feel like it was emerging, not sitting there.
- 5. Yes—I was serious, because I needed a way out that didn’t involve begging collectors, and ping-tennis looked like a door I could force open.

Q1: Among your friends in Paris, who truly helped you?

Q2: Did you ever entrust important works to friends or institutions for safekeeping?

Q3: What were the actual terms of your collaboration with Henri-Pierre Roché, and what financial exchanges occurred between you?

Q4: How did you obtain mirror materials, how did you paint on them, and why did you choose this medium?

Q5: Was promoting “ping-pong tennis” in New York a serious career shift?

Figure 6: Divergent “digital Sanyu” responses across participants to a shared set of five questions (A–H)

3 INTERACTION AND EXPERIENCE

3.1 Workshop

To examine Sanyu Studio’s interaction experience among art-interested users, we conducted a seven-day offline workshop at an art university in China. Participants were eight senior oil-painting undergraduates: four male and four female, aged

22 $\pm$ 1.20 years, labeled A–H. All signed informed consent forms for participation and anonymous publication. The workshop examined whether participants could generate diverse art-historical narratives from the same initial conditions.

Before the workshop, participants completed the short Need for Closure Scale (NFCS) [11] (Table 1,2) to measure their preference for certainty and rapid judgment under uncertainty. We compared NFCS scores with the final parameters of their generated Sanyu agents. Given the small sample, this comparison is treated as exploratory and does not support strong statistical claims.

Table 1: Operational Definitions of the Measured Variables

3.1 Category	Description
NFCS_total_score	calculated by summing all item scores after reverse coding; higher scores indicate a stronger preference for certainty and faster judgment.
session_count	total number of roundtable sessions initiated by the participant.
total_user_turns	total number of participant utterances across all sessions, with each user input counted as one turn.
avg_question_length	average length of participant utterances per turn.
avg_answer_length	average length of system responses per turn.
followup_ratio	proportion of utterances that continue the immediately preceding topic among all comparable turns.
micro_topic_count	total number of fine-grained topic units identified across all dialogues.
switch_rate	proportion of topic shifts between adjacent comparable turns.
mean_run_length	average number of consecutive turns spent on the same topic.
revisit_rate	proportion of turns that return to a previously abandoned topic.

Table 2: Summary Statistics of Participant Interaction Behavior and NFCS Scores

ID	A	B	C	D	E	F	G	H
NFCS_total_score	62	55	34	68	38	29	42	47
session_count	56	58	69	53	67	72	64	62
total_user_turns	401	428	536	382	517	563	489	462
avg_question_length	26.8	29.4	39.5	24.1	37.8	41.3	35.2	33.1
avg_answer_length	165.4	171.2	191.7	158.6	186.4	196.2	182.1	178.5
followup_ratio	0.661	0.586	0.389	0.714	0.421	0.351	0.462	0.510
micro_topic_count	74	86	127	67	119	139	108	99
switch_rate	0.339	0.414	0.612	0.286	0.578	0.648	0.535	0.490
mean_run_length	2.320	2.030	1.440	2.720	1.540	1.330	1.670	1.790
revisit_rate	0.20	0.22	0.30	0.17	0.29	0.32	0.27	0.25

Each participant completed at least 50 roundtable discussions, including at least five based on preset questions addressing disputed or under-documented issues in Sanyu studies (Figure 6):

Q1: Among your friends in Paris, who truly helped you?

Q2: Did you ever entrust important works to friends or institutions for safekeeping?

Q3: What were the actual terms of your collaboration with Henri-Pierre Roché, and what financial exchanges occurred between you?

Q4: How did you obtain mirror materials, how did you paint on them, and why did you choose this medium?

Q5: Was promoting “ping-pong tennis” in New York a serious career shift?

After brief training, all participants began from the same system state and evidentiary base, then freely explored related topics over seven days. We recorded dialogue content, retrieval context, and memories written into the Sanyu agent. On

the final day, we exported each participant’s final answers to the five questions and analyzed them with NFCS scores, interaction rounds, and question paths to examine how cognitive tendency shaped the generated “digital Sanyu.” (Table 1,2)

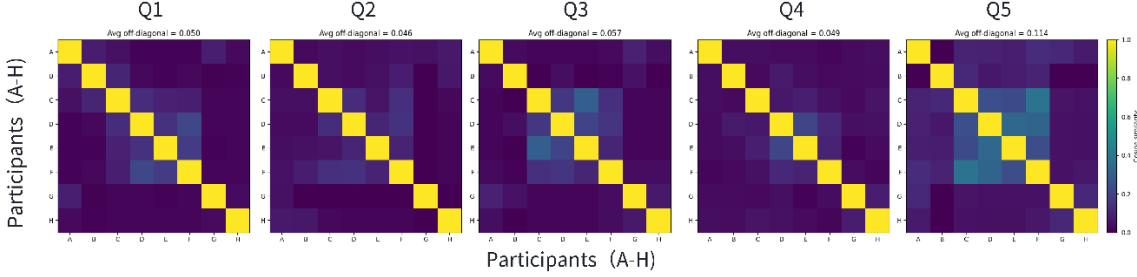


Figure 7: (A–H) Pairwise semantic similarity across eight participants’ final answers to five standardized questions (cosine similarity in an LSA space from TF–IDF; stopwords removal; length normalization).

3.2 Results

The results show that participants’ cognitive tendencies and question paths shaped the generated Sanyu agents, especially on issues with limited evidence or no clear consensus.

We extracted the eight agents’ final answers to the five preset questions and calculated pairwise semantic similarity. Off-diagonal mean similarity was low across all questions: Q1 = 0.050, Q2 = 0.046, Q3 = 0.057, Q4 = 0.049, and Q5 = 0.114, indicating broad narrative divergence. Although Q5 showed slightly higher consistency, no stable consensus emerged (Figure 7).

Memory logs further suggest that participants with higher NFCS scores converged faster, with fewer rounds, shorter questions, more follow-ups, and fewer topic shifts. Those with lower scores explored more openly, with more rounds, longer questions, and more frequent shifts and returns.

Overall, Sanyu Studio generated differentiated art-historical narratives as users varied their entry points and evidence-integration paths, supporting plural constructions of artistic meaning.

4 DISCUSSION

On the final day, we held an open discussion on user experience and future work (Figure 8). All participants agreed that Sanyu Studio offers art-history enthusiasts an accessible entry into Sanyu studies and that the roundtable format effectively simulates the comparison and negotiation of materials, evidence, and interpretive paths.

Participants also identified two limitations. First, although the system labels source status and separates factual materials from inferences, it may still over-infer when addressing personal experience, private relationships, or sensitive identity, raising ethical and privacy risks. Future work should add stricter content boundaries and human review. Second, the workshop included only eight art-university students with basic art-historical knowledge. Future studies will recruit broader participants to test usability among users without prior training.

5 REFLECTION AND CONCLUSION

5.1 Reflection

From an art-historical perspective, new technologies operate as both communication media and technical storage media, reshaping how artworks are preserved, circulated, and interpreted [2]. Concerns over AI-driven homogenization should therefore be understood within broader shifts in art-historical methods and knowledge structures. Although generative AI cannot yet produce original art-historical scholarship autonomously, LLMs can already support evidence organization, interpretive negotiation, and narrative generation [13, 14]. They may also serve as methodological intermediaries, helping broader audiences enter art-historical discussion and expanding the public reach of art-historical research.



Figure 8: Introducing the basic operating procedures of “Sanyu Studio” to oil painting students at an art academy.

5.2 Conclusion and Contributions

This paper introduced Sanyu Studio, a multi-agent roundtable system for art-historical narrative construction. We described its painting-agent design, evidence-organization mechanism, roundtable negotiation process, and human-AI workshop results.

Using Sanyu as a case, this paper explores how LLMs can organize artwork evidence, negotiate interpretive paths, and generate revisable artist narratives. By proceduralizing key research steps, Sanyu Studio lowers the threshold of art-historical inquiry and offers nonexpert art-history enthusiasts an interactive entry point for interpretation and narrative revision.

More broadly, this paper maps historical scholarship—question formation, evidence retrieval, multi-source negotiation, and narrative synthesis—onto a computable multi-agent mechanism. This approach can extend to historical figures, cultural heritage, event history, and other domains requiring narrative generation from multi-source evidence.

REFERENCES

- [1] Anderson, B.R. et al. 2024. Homogenization Effects of Large Language Models on Human Creative Ideation. *Creativity and Cognition* (Chicago IL USA, June 2024), 413–425.
- [2] Benjamin, W. et al. 2008. *The work of art in the age of its technological reproducibility, and other writings on media*. Harvard University Press.
- [3] Bin, Y. et al. 2024. GalleryGPT: Analyzing Paintings with Large Multimodal Models. (2024). <https://doi.org/10.48550/ARXIV.2408.00491>.
- [4] DaCosta, B. 2025. Speaking with the Past: Constructing AI-Generated Historical Characters for Cultural Heritage and Learning. *Heritage*. 8, 9 (Sept. 2025), 387. <https://doi.org/10.3390/heritage8090387>.
- [5] Ginsburgh, V. and Weyers, S. 2010. On The Formation of Canons: The Dynamics of Narratives in Art History. *Empirical Studies of the Arts*. 28, 1 (Jan. 2010), 37–72. <https://doi.org/10.2190/EM.28.1.d>.
- [6] Ginzburg, C. ed. 2013. *Clues, myths, and the historical method*. Johns Hopkins University Press.
- [7] Ginzburg, C. 1980. Morelli, Freud and Sherlock Holmes: Clues and Scientific Method*. *History Workshop Journal*. 9, 1 (Mar. 1980), 5–36. <https://doi.org/10.1093/hwj/9.1.5>.
- [8] Hilberg, R. 1989. *Legend, Myth, and Magic in the Image of the Artist*. Yale University Press.
- [9] Huang, L. et al. 2026. Artificial intelligence-based image interpretation system: Unlocking a new dimension of artwork analysis. *Array*. 29, (Mar. 2026), 100670. <https://doi.org/10.1016/j.array.2025.100670>.
- [10] Natale, S. et al. 2025. ChatGPT for cultural heritage and the customization of generative AI: A talkthrough analysis of the Luigi Einaudi chatbot. *New Media & Society*. (Oct. 2025), 14614448251384258. <https://doi.org/10.1177/14614448251384258>.
- [11] Roets, A. and Van Hiel, A. 2011. Item selection and validation of a brief, 15-item version of the Need for Closure Scale. *Personality and Individual Differences*. 50, 1 (Jan. 2011), 90–94. <https://doi.org/10.1016/j.paid.2010.09.004>.
- [12] Trouillot, M.-R. 2011. *Silencing the past: power and the production of history*. Beacon Press.
- [13] Um, N. 2024. Scholarly Writing in the Face of Generative AI: A View from Art History. *Ars Orientalis*. 54, 0 (Dec. 2024). <https://doi.org/10.3998/ars.7035>.
- [14] Wang, S. et al. 2025. ArtRAG: Retrieval-Augmented Generation with Structured Context for Visual Art Understanding. *Proceedings of the 33rd ACM International Conference on Multimedia* (Dublin Ireland, Oct. 2025), 6700–6709.
- [15] Wasielewski, A. 2023. *Computational Formalism: Art History and Machine Learning*. The MIT Press.
- [16] White, H.V. 2003. *Tropics of discourse: essays in cultural criticism*. Johns Hopkins Univ. Pr.
- [17] Wölfflin, H. et al. 2015. *Principles of art history: the problem of the development of style in early modern art*. Getty Research Institute.
- [18] Yan, Y. 2025. The Study on Sanyu’s Painting from the Perspective of the Fusion of Chinese and Western Aesthetics. *Proceedings of the 2025 4th International Conference on Social Sciences and Humanities and Arts (SSHA 2025)*. S. Jing et al., eds. Atlantis Press SARL. 167–179.
- [19] Zhang, W. et al. 2025. CultiVerse: Towards Cross-Cultural Understanding for Paintings with Large Language Model. *Proceedings of the 33rd ACM International Conference on Multimedia* (Dublin Ireland, Oct. 2025), 6710–6719.